

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 7: Meteorological Planning Charts (WITEM & SIGWX)

Syllabus Source: CAE Oxford (pp. 249-290) | Pass Target: >= 90% AviationExam

1. UPPER WIND & TEMPERATURE CHARTS (WITEM)

WITEM charts display forecast wind arrows and outside temperatures at specific flight levels (FL 050, 100, 180, 240, 300, 340, 390):

- Wind Arrow Decoding: Shaft points in the direction the wind is blowing. Barb markings:
Triangular pennant = 50 knots; Long barb = 10 knots; Short barb = 5 knots.
- Temperature Labels: At and above FL 240, temperatures are universally NEGATIVE; the minus sign is omitted! (e.g. '56' at FL 340 means -56°C!).

2. SIGNIFICANT WEATHER CHARTS (SIGWX)

High-level SIGWX charts (FL 250 to FL 630) show critical en-route flight hazards:

- Jet Streams: Represented by heavy black lines with arrows showing direction. Core speed: Double slash // indicates speed change of 20 kt. Hatching indicates CAT (Clear Air Turbulence) boundaries.
- Tropopause Heights: Printed in small rectangular boxes (e.g. '380' means tropopause is at FL 380). High tropopause = warm air mass (equatorial); Low tropopause = cold polar air mass.
- Cumulonimbus (CB): Identified as ISOL CB (isolated), OCNL CB (occasional), or FRQ CB (frequent) embedded in layer clouds.

3. COLD WEATHER ALTIMETER TEMPERATURE CORRECTIONS

In extreme cold temperatures, the barometric altimeter OVER-READS (indicates higher than true altitude!): 'High to low, look out below!'

- Master Cold Temperature Correction Formula: $\text{Correction (ft)} = 4 \text{ ft} \times (\text{Height above altimeter station} / 1,000 \text{ ft}) \times (\text{ISA Deviation in } ^\circ\text{C below standard})$

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Reading WITEM & SIGWX Charts

WITEM Decoding: - Pennant (solid triangle) = 50 kt. Long barb = 10 kt. Short barb = 5 kt. - Example: 2 pennants + 1 long barb = 50 + 50 + 10 = 110 knots. - Temperatures above FL100 omit minus signs: '56' means -56 C. SIGWX Jetstream Core: - The highest Clear Air Turbulence (CAT) is found on the COLD (polar) side and ABOVE the jet core, where horizontal and vertical wind shear is maximum!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Cold Weather Altimeter Correction on Final Approach

SCENARIO (Safety-Critical ICAO Winter Operation Drill): An aircraft is on an ILS approach to an airport in Scandinavia in winter: - Airport Elevation = 1,000 ft MSL - Outside Air Temperature (OAT) at the surface = -25°C - Decision Altitude (DA) on chart = 1,200 ft MSL (Height above runway = 200 ft) - Minimum Safe Altitude (MSA) = 4,000 ft MSL (Height above station = 3,000 ft) QUESTION: What is the true altitude of the aircraft if it levels at indicated MSA 4,000 ft? What indicated altitude must the pilot fly to maintain true MSA 4,000 ft? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate ISA Temperature at Airport Elevation (1,000 ft): - ISA at Sea Level = +15°C. - ISA lapse rate = -2°C per 1,000 ft. - ISA at 1,000 ft = +15°C - 2°C = +13°C. Step 2: Calculate ISA Deviation (Delta ISA): - Actual OAT = -25°C. - Delta ISA = Actual OAT - ISA = -25°C - (+13°C) = -38°C (38°C COLDER than standard!). Step 3: Calculate Altimeter Error at MSA (Height above station = 4,000 - 1,000 = 3,000 ft): - Rule: 4 ft per 1,000 ft of height per °C of cold deviation. - Correction = 4 ft x (3,000 ft / 1,000 ft) x 38°C - Correction = 4 ft x 3 x 38 = 12 x 38 = 456 ft error! Step 4: Determine True Altitude vs Required Indicated Altitude: - Because the air is cold, the altimeter over-reads by 456 ft! If flying at indicated 4,000 ft, TRUE ALTITUDE IS ONLY 3,544 ft (456 ft dangerously closer to mountains!). - To guarantee 4,000 ft of true clearance, the pilot MUST ADD 456 ft to the altimeter! - Minimum Indicated Altitude to fly = 4,000 ft + 456 ft = 4,456 ft! FINAL ANSWER: True altitude is 3,544 ft without correction. Pilot must fly indicated 4,460 ft.

[!] EXAM TRAP (AVIATIONEXAM) - The Cold Weather Altimeter Trap

- Cold air contracts the atmospheric column: True altitude is LOWER than indicated altitude! - When OAT is below 0°C, temperature corrections MUST BE ADDED to published altitudes (DA, MDA, MSA) so you don't hit the ground early!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: On an upper wind WINTEN chart at FL 340, a temperature is labeled '48' next to a wind arrow. What is the actual temperature?

- [A]: +48°C
- [B]: -48°C (minus sign is universally omitted at high flight levels)
- [C]: 48 Kelvin
- [D]: 48% relative humidity

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Above FL 240 in mid-latitudes, temperatures are permanently below zero. To reduce chart clutter, the negative minus sign is omitted.

Q2: In extremely cold weather (OAT substantially below ISA), how does an uncorrected pressure altimeter read?

- [A]: It reads lower than true altitude
- [B]: It reads HIGHER than true altitude (the aircraft is physically lower than the altimeter indicates!)
- [C]: It reads accurately
- [D]: It fails completely

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Cold air is dense, causing pressure levels to compress closer to the ground. The altimeter reads higher than actual true altitude ('high to low, look out below').

Q3: On a high-level SIGWX chart, what does a heavy black line with an arrow and a double slash // indicate?

- [A]: A thunderstorm front
- [B]: A JET STREAM, with the double slash indicating a speed change of 20 knots along the jet axis
- [C]: An oceanic boundary
- [D]: Volcanic ash

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Jet stream cores are depicted by heavy arrows. A double bar // represents a change in core wind speed of 20 kt.